D3.7. In each of the following parts, find a numerical value for div $\mathbf{D}$ at the point specified: (*a*) $\mathbf{D} = (2xyz - y^2)\mathbf{a}_x + (x^2z - 2xy)\mathbf{a}_y + x^2y\mathbf{a}_z$ C/m^2 at $P_A(2, 3, -1)$; (*b*) $\mathbf{D} = 2\rho z^2 \sin^2\phi\,\mathbf{a}_\rho + \rho z^2 \sin 2\phi\,\mathbf{a}_\phi + 2\rho^2 z \sin^2\phi\,\mathbf{a}_z$ C/m^2 at $P_B(\rho = 2, \phi = 110°, z = -1)$; (*c*) $\mathbf{D} = 2r \sin\theta \cos\phi\,\mathbf{a}_r + r\cos\theta\cos\phi\,\mathbf{a}_\theta - r\sin\phi\,\mathbf{a}_\phi$ C/m^2 at $P_C(r = 1.5, \theta = 30°, \phi = 50°)$.

Ans. (*a*) −10.00; (*b*) 9.06; (*c*) 1.29

Divergence

$$\text{div}\,\mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$ Rectangular

$$\text{div}\,\mathbf{D} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$ Cylindrical

$$\text{div}\,\mathbf{D} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial \phi}$$ Spherical

D3.9. Given the field $\mathbf{D} = 6\rho \sin\frac{1}{2}\phi\, \mathbf{a}_\rho + 1.5\rho \cos\frac{1}{2}\phi\, \mathbf{a}_\phi$ C/m^2, evaluate both sides of the divergence theorem for the region bounded by $\rho = 2$, $\phi = 0$, $\phi = \pi$, $z = 0$, and $z = 5$.

Ans. 225; 225

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D}\, dv$$

Energy and potential

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Energy Expended in Moving a Point Charge in an Electric Field

- Suppose we wish to move a charge Q a distance dL in an electric field E, the force on Q arising from the electric field is: $\mathbf{F}_E = Q\mathbf{E}$
- The component of this force in the direction dL ,which must be overcome is:

$$F_{EL} = \mathbf{F}_E \cdot \mathbf{a}_L = Q\mathbf{E} \cdot \mathbf{a}_L$$

- The force that needs to be applied, must be equal and opposite to the force exerted by the field:

$$F_{\text{appl}} = -Q\mathbf{E} \cdot \mathbf{a}_L$$

- Differential work done by external source to move Q is equal to:

$$dW = -Q\mathbf{E} \cdot \mathbf{a}_L dL = -Q\mathbf{E} \cdot d\mathbf{L}$$

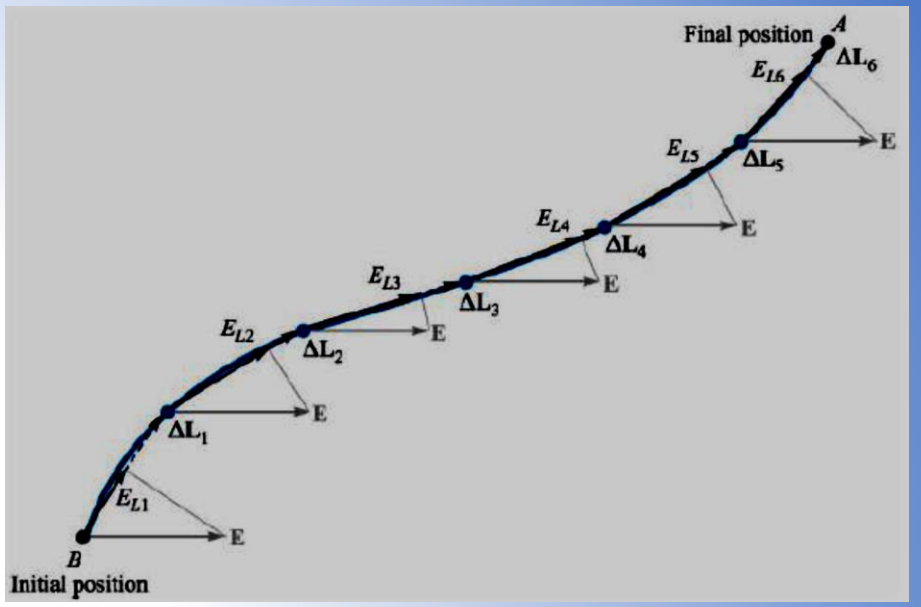
Energy Expended in Moving a Point Charge in an Electric Field

- The work required to move the charge a finite distance is determined by integration:

$$W = -Q\int_{\text{init}}^{\text{final}} \mathbf{E}\cdot d\mathbf{L}$$

- The path must be specified beforehand
- The charge is assumed to be at rest at both initial and final positions

The Line Integral



- The integral expression of previous equation is an example of a line integral, taking the form of integral along a prescribed path
- Without using vector notation, we can write:

$$W = -Q\int_{\text{init}}^{\text{final}} E_L dL$$

- The work involved in moving a charge Q from B to A is approximately:

$$W = -Q(E_{L1}\Delta L_1 + E_{L2}\Delta L_2 + \cdots + E_{L6}\Delta L_6)$$
$$W = -Q(\mathbf{E}_1 \cdot \Delta\mathbf{L}_1 + \mathbf{E}_2 \cdot \Delta\mathbf{L}_2 + \cdots + \mathbf{E}_6 \cdot \Delta\mathbf{L}_6)$$

The Line Integral

- If we assume that the electric field is uniform,

$$\mathbf{E}_1 = \mathbf{E}_2 = \cdots = \mathbf{E}_6$$

$$W = -Q\mathbf{E} \cdot \underbrace{(\Delta\mathbf{L}_1 + \Delta\mathbf{L}_2 + \cdots + \Delta\mathbf{L}_6)}_{\mathbf{L}_{BA}}$$

- Therefore,

$$W = -Q\mathbf{E} \cdot \mathbf{L}_{BA} \qquad \text{(uniform } \mathbf{E}\text{)}$$

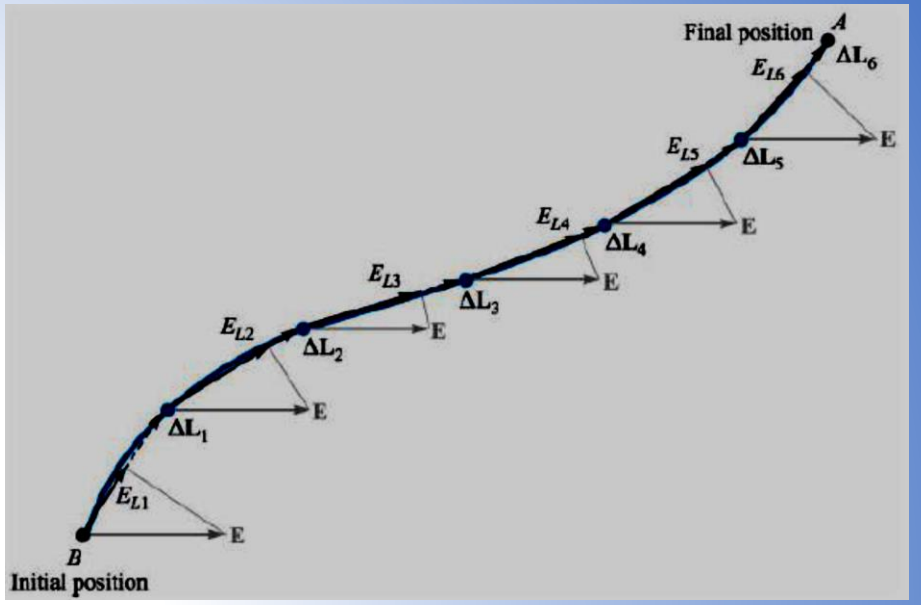
- Since the summation can be interpreted as a line integral, the exact result for the uniform field can be obtained as:

$$W = -Q\int_B^A \mathbf{E} \cdot d\mathbf{L}$$

$$W = -Q\mathbf{E} \cdot \int_B^A d\mathbf{L} \qquad \text{(uniform } \mathbf{E}\text{)}$$

$W = -Q\mathbf{E} \cdot \mathbf{L}_{BA}$ (uniform $\mathbf{E}$)

- For the case of uniform E, W does not depend on the particular path selected along which the charge is carried



■ **Example**

Given the non-uniform field $\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$, determine the work expended in carrying 2 C from $B(1,0,1)$ to $A(0.8,0.6,1)$ along the shorter arc of the circle $x^2 + y^2 = 1$, $z = 1$.

$d\mathbf{L} = dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z$ • Differential path, rectangular coordinate

$$W = -Q\int_B^A \mathbf{E}\cdot d\mathbf{L}$$

$$= -Q\int_B^A (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z)\cdot(dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z)$$

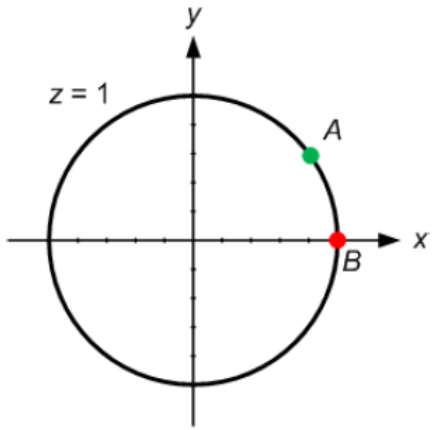
$$= -2\int_1^{0.8} y\,dx - 2\int_0^{0.6} x\,dy - 2\int_1^1 2\,dz$$

• Circle equation: $x^2 + y^2 = 1$

$$x = \sqrt{1-y^2}$$

$$y = \sqrt{1-x^2}$$

$$W = -2\int_1^{0.8} \sqrt{1-x^2}\,dx - 2\int_0^{0.6} \sqrt{1-y^2}\,dy - 2\int_1^1 2\,dz$$



$$= -2\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x\right]_1^{0.8} - 2\left[\frac{y}{2}\sqrt{1-y^2} + \frac{1}{2}\sin^{-1}y\right]_0^{0.6}$$

$$= \underline{\underline{-0.962\ \text{J}}}$$

$$\int \sqrt{a^2-u^2}\,du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\sin^{-1}\frac{u}{a}$$


Thank
you